BioLab - application for online analysis using Lab Streaming Layer for education and research purpose

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***Abstract*—In this paper, we present BioLab, an educational and research application designed for real-time streaming and analysis of biomedical signals using the Lab Streaming Layer (LSL). BioLab facilitates the streaming, visualization, and storage of signals such as electrocardiography (ECG), electroencephalography (EEG) and electromyography (EMG), among others, from various biomedical devices, which can then be analyzed in real-time or post-acquisition. We have developed specific scripts for ECG and EMG data acquisition, along with MATLAB scripts to plot, visualize, and filter the received signals. The system's configuration is easily adjustable through JSON configuration files, allowing flexibility in the stream settings. While our current focus is on real-time visualization and basic noise filtering, we discuss the potential for future classification of cardiovascular diseases based on the spectrograms generated from the streamed ECG data. This platform aims to serve as an essential tool for researchers and educators in the field of biomedical engineering, providing a flexible environment for the analysis of both raw and processed biosignal data.**

***Index Terms*—bioelectric signals, signal processing, biomedical applications**

I. INTRODUCTION

The increasing availability of biomedical data through wearable sensors and medical devices has led to a growing need for real-time signal analysis platforms. BioLab is an educational and research application developed to meet this need by facilitating the acquisition, streaming, and visualization of biomedical signals. By leveraging the Lab Streaming Layer (LSL) protocol, BioLab allows multiple devices to stream various biosignals, such as electrocardiogrpahy (ECG), electromyography (EMG) or electroencephalography (EEG), that can be further analyzed in real-time or stored for post-processing.[1]

Real-time analysis of biosignals can have profound applications in both educational and research environments, providing timely insights into physiological conditions. Although this project currently focuses on data acquisition and basic visualization, the future development of machine learning models to classify biomedical conditions—particularly cardiovascular diseases, as we have worked with them—using this software has significant potential. In this paper, we describe the system architecture of BioLab, its configuration options, the development of MATLAB scripts for signal analysis, and preliminary findings in the potential classification of ECG abnormalities.[2]

II. BIOMEDICAL SIGNALS

Biomedical signals are electrical potentials that reflect the activity of biological systems, commonly used in the field of healthcare and biomedical engineering. These signals provide valuable insights into various physiological functions, such as heart rhythm, muscle activity, and brain function. Through non-invasive methods, biomedical signals can be captured in real-time, allowing for both immediate and long-term monitoring of a patient's condition. In the context of our application, BioLab has been tested primarily with three types of biomedical signals - electrocardiography, electromyography, electroencephalography.[2]

A. Electrocardiography

ECG signals represent the electrical activity of the heart over time. The heart generates electrical impulses that cause contractions, and these impulses can be measured through electrodes placed on the skin. An ECG trace typically shows the repeating cycle of heartbeats, characterized by specific waveforms: the P wave, QRS complex, and T wave. The amplitude of this signal goes from 0.1 to 5 mV, depending on the placement of the electrodes and the patient's physiology. Most of the useful information is concentrated below 100 Hz, with typical heartbeats generating frequencies around 0.67 to 4 Hz (for heart rates of 40-240 bpm).[3]

Basic electrode placement consist of 3 electrodes (named also 3-lead connection) - placed on:

- **RA** - left top side of chest, near left shoulder,
- **LA** - right top side of chest, near right shoulder,
- **LL** - left center part of chest, near left rib cage.

3-lead ECG connection (figure 1) is commonly used for diagnosing heart conditions such as arrhythmias, myocardial infarctions, or BPM evaluations.

More sophisticated placement - aslo named as 12-lead connection, consist of connection of 10 electrodes:

- **RA** - left forearm or wrist,
- **LA** - right forearm or wrist,
- **RL** - right lower leg or anckle,

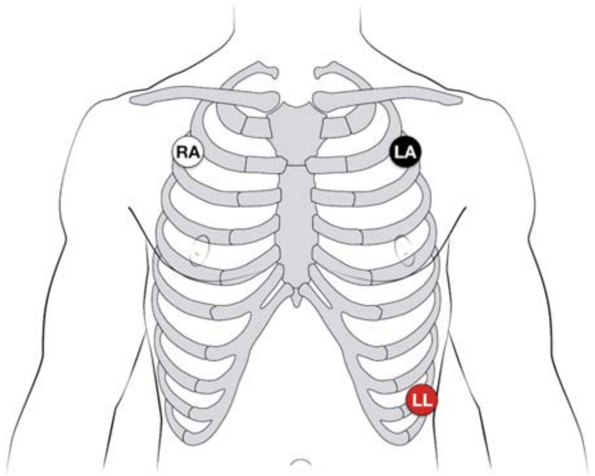

Fig. 1. 3 lead connection for ECG measurement

- **LL** - left lower leg or anckle,
- **V1 - V6** - placement showed on figure 2.

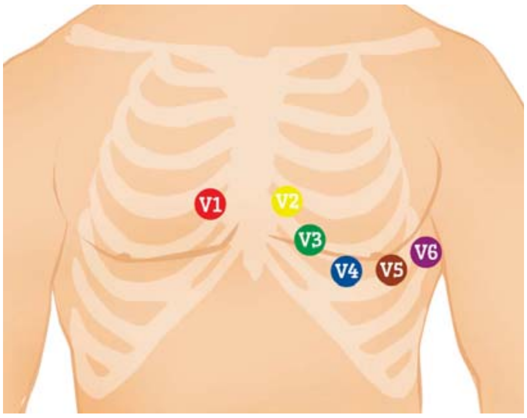

Fig. 2. 12-lead connection for ECG

Consequently - by 12 lead ECG are analysed 12 waveforms:

- **Lead I** = $LA - RA$
- **Lead II** = $LL - RA$
- **Lead III** = $LL - LA$
- **aVR** = $RA - \frac{LA+LL}{2}$
- **aVL** = $LA - \frac{LL+RA}{2}$
- **aVF** = $LL - \frac{LA+RA}{2}$
- **V1** = $V1 - \frac{LA+LL+RA}{3}$
- **V2** = $V2 - \frac{LA+LL+RA}{3}$
- **V3** = $V3 - \frac{LA+LL+RA}{3}$
- **V4** = $V4 - \frac{LA+LL+RA}{3}$
- **V5** = $V5 - \frac{LA+LL+RA}{3}$
- **V6** = $V6 - \frac{LA+LL+RA}{3}$

12 lead ECG is used mostly for classification of more complex heart diseases, where 3 lead measurement is not enough.

B. Electromyography

EMG signals reflect the electrical activity of muscles. When a muscle contracts, electrical potentials are generated, which can be measured through electrodes either placed on the skin or inserted into the muscle. EMG is widely used to assess muscle function and detect neuromuscular diseases. The amplitude of EMG signals can range from 0.01 to 5 mV, depending on the muscle size and electrode placement and they typically lie between 10 Hz and 500 Hz, with most of the energy concentrated between 50 Hz and 150 Hz.[4]

EMG is often used in diagnostics for conditions like muscular dystrophy, nerve damage, and to monitor rehabilitation progress after injuries.

C. Electroencephalography

EEG signals capture the electrical activity of the brain, recorded by placing electrodes on the scalp. These signals are crucial for understanding brain functions and diagnosing neurological disorders. EEG measures the brain's spontaneous electrical activity over a period of time. Typically, EEG signals have very low amplitudes, ranging from 10 to 100 µV, making them much smaller than ECG or EMG signals. EEG signals are categorized into different frequency bands:[5]

- **Delta (0.5-4 Hz)**: Associated with deep sleep.
- **Theta (4-8 Hz)**: Linked to drowsiness or early sleep.
- **Alpha (8-12 Hz)**: Reflects a relaxed state, often seen with closed eyes.
- **Beta (12-30 Hz)**: Related to active thinking or concentration.
- **Gamma (30-100 Hz)**: Associated with higher-level cognitive functions.

Standard placement of electrodes is defined as 10-20 standard (figure 3) - where

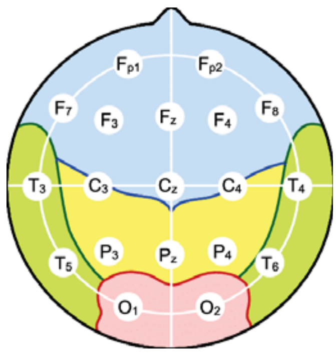

Fig. 3. 10-20 standard placement in EEG monitoring

In case of further analysis, EEG are key measurements for brain-computer interface (BCI) systems. These BCI systems can be divided to two groups - active and passive BCI.

1) Passive BCI: In passive BCI systems is no need of actions, interactions of human. These systems are commonly used for classification of diseases, rehabilitation processes. Passive BCI is also used for prediction of abnormalities of behavior that could lead to disease in future of human life.[6]

2) Active BCI: Active BCI is mostly used in systems where is need of human actions. It can be divided to mental counting, motor imagery, music imagery and mental arithmetic. These classes of active BCI are mostly used to control systems because of assumption of higher final efficiency.[6]

III. System architecture

BioLab application architecture is buiilded on Lab Streaming Layer library - often used for streaming and synchronizing multiple signals with different parameters - such as sampling frequency. This component also allows real-time analysis - such as filtering, frequency analysis and further classification or prediction of events in real time. LSL also allows multiple devices - such as computers connect to stream and process.

Architecture of application is shown on fig. 4, where multiple devices with different sampling frequency (depending on sampled signal) are connected to BioLab server, and further analysis is implemented on Computer 1 to n. This computers must be connected to the same internet network. BioLab application also includes open-source datasets that can be streamed and processed as example.

BioLab application consists of several modular components designed to work together seamlessly for biomedical signal acquisition, streaming, and analysis. The system is structured into four main sections:

- configuration files,
- streaming scripts,
- logs,
- main execution task.

A. *Configuration files*

Configuration files are stored in JSON format. These files contain the parameters for each stream, such as stream name, sampling frequency, number of channels, and data type. Each configuration file is specific to the signal type and the device being used. The structure of these files allows users to easily modify or add new streams by simply updating data in the JSON file.

B. *Streaming scripts*

The core functionality of BioLab revolves around the streaming scripts. These scripts, developed in Python, use the Lab Streaming Layer (LSL) to stream data over a network. Separate scripts have been developed for different signal types. For now, the scripts for ECG and EMG signals streaming connects to the respective device, retrieve the data in real-time, and stream it using LSL, where it can be received by MATLAB or other compatible software in any other device that supports it.

C. *Logs*

A dedicated log system records all streamed data for auditing and post-analysis purposes. Each stream generates a log file, which tracks the data sent from the device, ensuring that the signals are recorded and available for later use.

D. *Main execution task*

The execution of the BioLab system is orchestrated by a central main function. This method is responsible for initializing and managing all active streams based on the configuration files provided. When the system is started, the main method reads the available JSON configuration files and creates separate subprocesses for each signal stream.

The main method performs several key tasks:

- **Configuration loading**: Firstly loads all configuration files from the core directory, ensuring that each stream's parameters (e.g., sampling rate, number of channels) are correctly initialized.
- **Stream Initialization**: Secondly, each stream described in the configuration files will be initialized in the appropriate Python script.
- **Process Management**: The method monitors the status of each streaming process. If any process unexpectedly terminates, the main method is designed to restart the stream, ensuring continuous data collection.
- **Logging and Monitoring**: While the streams are active, the main method writes log entries, detailing the state of each stream and any potential errors. This log information is crucial for debugging and ensures transparency in the streaming operations.

The main function acts as a central controller, allowing the system to start and manage multiple biomedical streams simultaneously, ensuring that the user can receive real-time data from several devices without manual intervention. It is also designed to be scalable, allowing additional streams to be added by updating the JSON configuration files and without requiring changes to the underlying architecture.

IV. Signal acquisition and processing

BioLab supports the acquisition of any type of biomedical signal. It has been tested with ECG and EMG signals through custom Python scripts that interact with the devices via Bluetooth. These scripts connect to specific sensors, retrieve the raw biosignal data, and push the data to an LSL outlet.

As an example of processing, we have implemented MATLAB scripts to receive, visualize, and filter the ECG signals in real-time. The ECG postprocessing scripts allow for two types of visualizations:

- **Raw Signal Plot**: Displays the unfiltered ECG signal as received from the device in real time.
- **Filtered data and further time-frequency analysis**: The ECG signal is filtered to remove baseline noise (at 30 Hz and 60 Hz), then visualized as a spectrogram. This provides a time-frequency-based representation of the signal, highlighting abnormal patterns and making it easier to identify.

Similarly, EMG signals are processed and streamed, allowing for real-time muscle activity analysis. MATLAB scripts have been developed to handle both single and multi-channel EMG data streams.

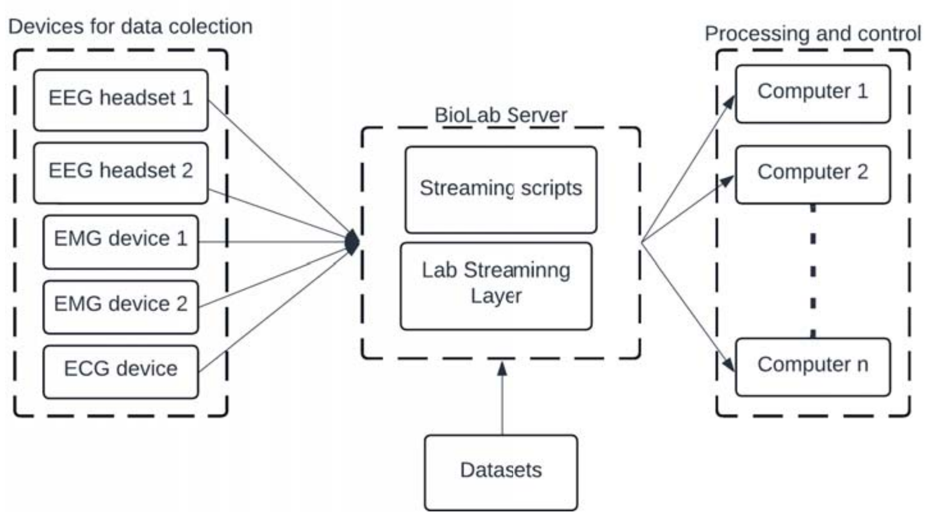

Fig. 4. BioLab application architecture

V. FUTURE LEADINGS

While BioLab currently focuses on real-time signal acquisition and visualization, there is significant potential for extending its capabilities to disease classification. We have experimented with an ECG dataset containing signals from 16 different cardiovascular conditions. By visualizing the signals as spectrograms, distinct patterns emerged that could be leveraged to classify diseases such as arrhythmias, tachycardias, and other abnormal heart rhythms.[7]

A. Potential for Machine Learning-Based Classification

Machine learning, particularly deep learning techniques like convolutional neural networks (CNNs), has proven to be highly effective for the classification of time-series data, including biomedical signals such as ECG. Spectrograms, which transform time-domain signals into the time-frequency domain, offer a rich source of information for pattern recognition. In the context of ECG, spectrograms can highlight frequency shifts, irregular rhythms, and anomalies that are often associated with specific cardiac conditions.[8][9]

For instance, arrhythmias often present as irregular heartbeats that produce idistinctive frequency patterns when transformed into spectrograms. Tachycardia, characterized by an abnormally fast heart rate, can also be identified through the concentration of high-frequency components. Given these observable differences in the spectrogram representation, CNNs can be trained to automatically recognize and classify these patterns with high accuracy.[8][9]

B. Machine Learning Workflow for ECG Classification

In future iterations of BioLab, the following machine learning workflow could be implemented:

- **Data Collection**: The biosignals streamed by BioLab would be collected and preprocessed, with spectrograms being generated from the raw time-series data.
- **Feature Extraction**: Using the spectrogram images, relevant features such as frequency bands, rhythm regularity, and anomaly markers can be extracted. This step often involves techniques like Short-Time Discrete Fourier Transform (STDFT) to highlight time-varying frequency content.
- **Training a CNN Model**: A CNN would be trained on a labeled dataset of ECG spectrograms, where each spectrogram is tagged with its corresponding cardiovascular condition. This training phase involves optimizing the model to learn patterns in the spectrograms that differentiate between diseases.
- **Validation and Testing**: The trained model would be validated and tested on unseen data to assess its ability to generalize and accurately classify new ECG signals.
- **Integration into BioLab**: Once trained, the CNN model can be integrated into BioLab to provide real-time classification as an example. As the ECG signal is streamed, the system would generate spectrograms and feed them into the CNN for immediate disease detection.

C. Challenges and Future Work

While the use of machine learning for ECG disease classification is promising, there are several challenges to consider. These include the need for a large, diverse, and labeled dataset for training the model, the computational complexity of real-time processing, and the integration of these systems into clinical workflows. Moreover, different types of noise in the ECG signals, such as baseline drift or power line interference, can affect classification accuracy, making it essential to implement robust signal preprocessing techniques.[10]

Nevertheless, BioLab lays a strong foundation for the future incorporation of machine learning algorithms and real-time signal processing of biomedical data.

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VI. Applications and Use Cases

BioLab has multiple applications in the fields of education and research, including:

- **Real-time Monitoring**: Researchers can monitor ECG and EMG signals in real-time, observing physiological changes as they occur.
- **Post-processing Analysis**: Data streamed and logged by BioLab can be analyzed later using MATLAB or other signal processing software.
- **Educational Tool**: The platform can be used to teach students about biomedical signal acquisition, processing, and analysis. By allowing real-time interaction with the data, it enhances the learning experience.
- **Disease Research**: With further development, BioLab could be used as a tool for disease research, enabling the classification of conditions from biosigals data streams.

VII. Future Work

Future developments of BioLab can be focus on the integration of machine learning techniques to classify diseases based on ECG and EMG signals. We aim to:

- Train convolutional neural networks (CNNs) using spectrograms of ECG signals.
- Expand the range of biomedical signals supported by the platform, including the scripts for EEG and respiratory data, p.e.
- Improve the real-time visualization interface for easier user interaction.
- Enhance filtering techniques to minimize noise interference in low-quality data streams.

VIII. Conclusion

BioLab application is a flexible, real-time signal streaming and analysis platform designed for educational and research purposes. Through the use of LSL, we have successfully developed a system that streams and processes biosignals in real-time, providing valuable tools for both immediate analysis and post-processing. The platform's modular design, combined with its integration with MATLAB for visualization, makes it highly adaptable for various biomedical signal applications. With future developments focused on disease classification, BioLab holds significant potential for enhancing research in human body activity monitoring.

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